

Research Article

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Extracting Inductive Bias Using Generative Models

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Abstract: Inductive biases shape how neural networks generalize from limited data, yet encoding them in network connectivity often relies on domain expertise. This paper presents a generative approach that draws on related tasks to discover shared structural constraints. For each task, a generative model approximates a family of hypothesis spaces that admit near-optimal solutions. Intersecting these task-specific families narrows the search to structures compatible with all tasks, automating the discovery of a shared inductive bias (want lossless). Experiments on synthetic tasks demonstrate improved data efficiency and generalization to unseen tasks, supporting the hypothesis that the recovered constraints capture transferable structure.

Keywords: inductive bias, generative model

1 Introduction

Choosing an appropriate inductive bias is crucial for generalization [1]. By constraining the hypothesis space to reflect task structure, it can enable effective learning from limited data. Examples include local connectivity and spatial weight sharing in convolutional networks [2], and recurrent state updates with parameters shared across time [3]. Suitable parameter-sharing patterns also enforce equivariance to task-relevant transformations [4].

Selecting suitable structural biases by hand requires domain expertise, particularly when the relevant properties of a new task are not known in advance [5]. This motivates learning such biases from data. For example, Zhou et al. [6] meta-learn layer reparameterizations to discover shared parameter structures, including symmetry-inducing patterns. Atanov et al. [7] learn generative priors from convolutional filters of previously trained networks and use them to guide learning on new datasets. Sun et al. [8] extract task-specific subnetworks by iterative pruning and share parameters where these subnetworks overlap. Ando and Zhang [9] learn a shared predictive structure across tasks while allowing task-specific predictors.

More generally, a single task may admit multiple near-optimal structures, and its data alone need not identify which constraints will transfer to other tasks. Baxter [1] provides a theoretical foundation for learning such constraints across an environment of related tasks. Under capacity and sampling assumptions, his bounds relate empirical performance across sufficiently many source tasks to the expected risk of the best task-specific hypotheses in a selected space on new tasks from the same environment, with high probability. Our focus is on extracting one transferable connectivity pattern from diverse trained solutions: a mask that remains useful when fresh weights are learned for a new task.

In this paper, we present a generative approach to learning a shared connectivity mask for related tasks. Building on generative modeling of trained networks [7, 10], we construct a bank of functional maps for each task from networks trained at different sparsity levels. Each map records neuron responses and connection sensitivities on fixed probe examples, together with the connectivity mask. Task-specific generators use these banks to propose new masks, extending the search beyond the stored solutions. A shared, task-conditioned evaluator predicts their loss and guides the search, following the principle of surrogate-guided architecture search [11].

The order of stored solutions and hidden neurons is arbitrary, so the generators process them as sets [12]. Their Transformer encoders omit positional encodings for these elements and average neuron representations to summarize each solution. An additional penalty encourages identical outputs when the bank is reordered while keeping the sampled noise fixed. An agreement loss encourages generators for different tasks to propose similar masks after aligning hidden neurons.

Search alternates mask generation with candidate evaluation on all source tasks: fresh weights are trained on support data, and loss is measured on separate query data. These measurements update the evaluator and generators, while selected trained models replenish the functional banks. Finally, a criterion based on the worst source-task performance selects one mask with a fixed connection budget using separate selection data. We freeze this mask before evaluating unseen tasks, learning weights independently for each task.

Our main contributions are as follows:

- We propose a generative framework for learning a shared connectivity mask as a structural inductive bias, with a fixed connection budget and independently trained task-specific weights.
- We introduce functional maps of trained networks as inputs to mask generation, combining neuron responses, connection sensitivities, and connectivity with an encoder invariant to solution and hidden-neuron ordering.
- We develop a cooperative search procedure that combines task-specific generators, agreement between their proposals, and a shared loss evaluator updated through candidate training on source tasks.
- We demonstrate improved data efficiency on previously unseen tasks using synthetic pattern-detection problems with known reference connectivity and image-based set-regression tasks, with comparisons against dense networks and matched-budget sparse baselines.

2 Related Works

TODO

3 Preliminary

3.1 Tasks and hypothesis spaces

We adopt the formulation of inductive bias learning introduced by Baxter [1]. Let $\mathcal{X}$ and $\mathcal{Y}$ denote the input and output spaces, and let $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, \infty)$ be a loss function. A supervised learning task is represented by a probability distribution P on $\mathcal{X} \times \mathcal{Y}$. A hypothesis is a function $h : \mathcal{X} \rightarrow \mathcal{Y}$, whose population risk is

$$R_P(h) := \mathbb{E}_{(x,y) \sim P} [\ell(h(x), y)]. \quad (1)$$

We assume that the expectations appearing below are well defined and finite.

For a single task, a task learner receives a sample $S = \{(x_j, y_j)\}_{j=1}^m$ drawn independently from P . Its empirical risk is

$$\hat{R}_S(h) := \frac{1}{m} \sum_{j=1}^m \ell(h(x_j), y_j). \quad (2)$$

A task learner $A_{\mathcal{H}}$ maps this sample to a hypothesis $A_{\mathcal{H}}(S) \in \mathcal{H}$ within a given hypothesis space $\mathcal{H}$. The choice of $\mathcal{H}$ encodes an inductive bias: restricting the available functions can facilitate generalization from limited data, provided that the space retains sufficiently good solutions.

3.2 Task environment and bias learner

Let Q be a probability distribution over a collection $\mathcal{P}$ of task distributions. Source tasks $P_1, \dots, P_n$ are drawn independently from Q , and each supplies a sample S_i of independent observations. Given a family of hypothesis spaces $\mathbb{H}$, a bias learner B uses these samples to select a shared hypothesis space:

$$\hat{\mathcal{H}} := B(S_1, \dots, S_n) \in \mathbb{H}, \quad (3)$$

Task learners use the selected space to learn their own hypotheses on new tasks drawn independently from Q . Following Baxter, the suitability of a space for this environment is measured by

$$R_Q(\mathcal{H}) := \mathbb{E}_{P \sim Q} \left[\inf_{h \in \mathcal{H}} R_P(h) \right]. \quad (4)$$

The task-specific infimum allows different hypotheses to solve different tasks within the same shared space.

3.3 Connectivity as a structural inductive bias

In our setting, hypothesis spaces are indexed by connectivity masks. Fix a network architecture with E candidate connections in the layer whose structure is to be learned. A binary mask

$$M \in \mathcal{M} \subseteq \{0, 1\}^E \quad (5)$$

specifies which connections are available. The admissible set $\mathcal{M}$ may impose a fixed connection budget or additional connectivity constraints.

Let $f_{w,M}$ denote a predictor with mask M and numerical parameters w . Masked connections are set to zero, while w includes the active weights and the remaining trainable parameters. Each mask defines a hypothesis space

$$\mathcal{H}_M := \{f_{w,M} : w \in \Theta_M\}, \quad \mathbb{H}_{\mathcal{M}} := \{\mathcal{H}_M : M \in \mathcal{M}\}, \quad (6)$$

where Θ_M is the admissible parameter set.

Learning a structural bias therefore amounts to selecting a mask whose induced space is suitable for the task environment. The corresponding population objective is

$$\inf_{M \in \mathcal{M}} R_Q(\mathcal{H}_M) = \inf_{M \in \mathcal{M}} \mathbb{E}_{P \sim Q} \left[\inf_{w \in \Theta_M} R_P(f_{w,M}) \right]. \quad (7)$$

In this setting, the bias learner B selects a mask $\hat{M}$ and thereby returns the space $\hat{\mathcal{H}} = \mathcal{H}_{\hat{M}}$. The selected mask is fixed across tasks, while a task learner trains numerical parameters from scratch for each new task. Connectivity constrains the support of the weights; equality constraints between active weights require an additional parameter-sharing mechanism.

3.4 Learning new tasks from finite samples

The environment risk measures the best solutions available within a hypothesis space. To describe learning from limited data, we also specify how a task learner uses that space.

For a fixed mask M , let A_M denote the task learner operating within $\mathcal{H}_M$. It returns a predictor $A_M(S) \in \mathcal{H}_M$ using sample S . Its expected risk after observing m labeled examples is

$$\mathcal{R}_m(M) := \mathbb{E}_{P \sim Q} \mathbb{E}_{S \sim P^m} [R_P(A_M(S))]. \quad (8)$$

For a mask $\hat{M}$ selected by the bias learner B , this quantity is evaluated conditional on the source data and the selected mask. The new task and its sample are independent of this selection.

Our objective is to learn a fixed structure that improves $\mathcal{R}_m$ at limited sample sizes, or reaches a prescribed risk level with fewer examples than comparison structures under a common training protocol. A low value of $R_Q(\mathcal{H}_M)$ establishes that good solutions are available. Whether the training procedure can obtain them efficiently from finite samples is a separate question.

4 Method

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5 Experiments

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6 Conclusion

TODO

References

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